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Implementation of penalized survival models in churn prediction of vehicle insurance

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ABSTRACT

The insurance industry plays an indispensable role in the financial structure of today's world. Vehicle insurance accounts for a large part of the income of insurance companies, and retaining regular clients has proved economical, making car insurance clients' churn management a priority. Meanwhile, clients' churn will determine companies' financial status and reflect the service and management level of the company. This paper proposes a new model, i.e., combined the Cox model with variable penalties (Lasso, SCAD, and MCP) to model the clients' churn problems and determine the crucial factors that affect clients' decisions, based on their personal information and behavior data, which are provided by a large insurance company in China. This model is proved to be successful in identifying the client churning and making comparisons among the penalties. The variable penalties model reveals the most important factor about clients' churn problems, which can provide a reliable basis for the product development of insurance companies. MCP creates the greatest sparsity but consequently loses some information. We suggest using SCAD in the model, as it balances the sparsity and information reservation. Furthermore, a new approach is proposed to building a dynamic threshold of churning probabilities, which companies can use to manage their clients more flexibly. In this way, clients who are prone to churn can be identified in advance, and the maintenance cost of client management can be reduced accordingly.

1. Introduction

With the continuous optimization of the economic structure, the insurance industry plays a more important role now than ever (Eling and Jia, 2019; Lee, Chang, Arouri, and Lee, 2016). The 2008 financial crisis led to the bankruptcy of a large number of companies around the world, and accordingly the default risk borne by the insurance industry has greatly increased (Ashraf, Zheng, Jiang, and Qian, 2020). During this period, the insurance industry experienced negative growth across the board, and reached its trough in 2009. However, this situation has been changed in 2010, one of the main reasons is that it has received policy support from various countries, and insurance companies are constantly exploring and innovating. With the recovery of the world economy, especially in recent years, the size of the global insurance market has reached a record high. In China, according to the State Council's new guidance for the insurance industry, by the end of 2020, the "insurance depth" (total premium income/GDP) will reach 5%, and the "insurance density" (total premium income/total population) will reach CNY 3500 per capita. Meanwhile, car insurance accounts for a large part of property insurance (in China, this proportion is 70%), and in many jurisdictions, it is compulsory to sign proper insurance to keep a car (Wanqing, 2020). Therefore, all major insurance companies have launched various types of vehicle insurance and invested funds to promote the vehicle insurance business, to penetrate markets, and keep existing clients.

Today the research on insurance industry mainly focused on pricing (Liu, Li, and Yuan, 2018; Verbelen, Antonio, and Claeskens, 2018). However, when modeling finance in reality, insurance companies will consider client churn prediction as an essential factor because it determines the future revenue and has a great impact on companies' financial status (Cooper, Gwin, and Wakefield, 2008; Ko, Kim, Kim, and Woo, 2008). A low client churn rate is a critical index that an insurance company will pursue. It mirrors clients' satisfaction and loyalty and also reflects the service and management level of the company. Clients' renewals will bring enormous benefits to a company in two ways: first, the cost of keeping a regular client is far lower than acquiring a new one, thus having clients renew their policies can reduce costs. Second,

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experience shows that insurance companies receive fewer claims from regular clients, which means lower payments. Nevertheless, existing research on the client churn problem focuses mainly on fields such as mobile communication and banking, and few focus on insurance, and insurance companies today usually manage clients based on their past experience (Cappiello, 2020). However, experience is a very broad and unquantifiable concept (Holmlund et al., 2020), and these decision-making methods that rely on past experience can lead to inevitable mistakes.

Therefore, it is the main purpose of this paper to select appropriate research methods to mine the information and quantifiable experience needed by insurance companies from massive customer data. In the past studies, many methods just pursued more accurate churn prediction. Most of these forecasting studies are based on single classification forecasting models, but the massive nature of big data greatly increases the difficulty. Therefore, this paper introduces the Cox model in the clients' churn problems, while incorporating variable penalty as an additional tool, which can model the clients' churn problem and identify the key factors that affect clients' decisions. The proposed model not only accurately predicts clients' churn rates, but also finds out the most important factor in clients' churn problems. A good predictive model can assign high churn probabilities to true churners. Therefore, by balancing the classification of precision and recall, this paper develops a dynamic threshold of client retention probability over time, which can help insurance companies maintain client relationship dynamically.

The rest of this paper is organized as follows. A literature review is given in Section 2. Section 3 introduces the Cox regression model, and Section 4 describes the variable selection. The simulation and applications are given in Section 5, and Section 6 concludes the article.

2. Literature review

Recent client management research can be divided into two major categories: The first follows the Customer Lifetime Value (CLV) (Berger and Nasr, 1998; Kotler, 1974). Keaveney and Parthasarathy (2001) studied customer value and came to the conclusion that retaining clients is the most crucial work of client management. With the rapid development of computing science, data mining techniques have inspired scholars to explore totally new directions in customer churn issues. Rygielski, Wang, and Yen (2002) applied Neural Networks to identify valuable customers. Bin (2005) predicted customers' churn rate based on statistical analysis, Neural Networks, and Genetic Algorithms. Support Vector Machines (SVMs) were also shown able to predict churn in subscription services (Coussement and Van den Poel, 2008), and SVMs outperformed traditional logistic models, whereas Random Forests outperformed SVMs. Xie, Li, Ngai, and Ying (2009) proposed a new learning method, called improved balanced Random Forests and showed that in churn prediction for banks, it produced better results than other Random Forests such as balanced and weighted Random Forests. Coussement and De Bock (2013) and Zhu, Baesens, and Broucke (2017) made a thorough comparison of various methods of churn prediction, including some ensemble solutions such as Bagging and AdaBoost. Customer churn was introduced into the telecommunication sector (Vijaya and Sivasankar, 2018), with rough set theory combined with ensemble classification techniques. De Caigny, Coussement, and De Bock (2018) proposed a hybrid algorithm based on logistic regression and Decision Trees to detect customer churn. Postigoboix and Melusmoreno (2018) used Agent-Based Modeling to detect potential churners for mobile network operators. Yu et al. (2018) solved this problem using an improved BP network with iterative particle classification optimization and particle fitness optimization. De Caigny, Coussement, De Bock, and Lessmann (2020) used benchmarking convolutional neural networks to forecast customer churn. Lalwani, Mishra, Chadha, and Sethi (2022) apply the most popular machine learning models to predict customer churn, and empirical results show that AdaBoost and XGboost Classifier have the highest accuracy. However, the above methods still have certain shortcomings, and these methods do not provide the ability of comprehensive prediction. First, these AI methods may not produce the expected results using new datasets. Second, these methods may suffer from "over-learning" problems. Finally, these methods generally suffer from data imbalance and noise.

In recent years, knowledge of survival analysis has also been applied in client management (Lee et al., 2018; To, Chau, and Kan, 2020). Survival analysis is mainly used in biological and medical research. It has advantages for dealing with censored data, and information from censored samples will not be wasted. Discrete survival models was used to predict credit card default (Djeundje and Crook, 2019). Li, Li, and Li (2019) studied credit card default problems using the Cox proportional hazards model. Hu, Chen, and Chen (2021) applied an extended Cox proportional hazards model to discover factors that contribute to customer churn in shared-cars field. Similar fields include traditional broadcast industry (Li et al., 2021) and telecommunication industry (Alboukaey, Joukhadar, and Ghneim, 2020; Amin et al., 2019). As for insurance fields, we see Cox regression model is proposed to analyze clients' repurchasing behavior with lifestyle segmentation (Ansell, Harrison, and Archibald, 2007). However, little research has been conducted on car insurance renewal problems. In the field of insurance, customer churn is one of the main factors affecting the profits of insurance companies and deserves careful study.

This paper proposes a new model, i.e., combined the Cox model with variable penalties (Lasso, SCAD, and MCP) to predict a client's leaving and distinguish regular clients for vehicle insurance, through knowledge of survival analysis. In essence, this can also be understood as a data mining technology, which aims to help insurance companies maintain client relationship dynamically and reduce the churn. We chose the Cox regression model, which is the best-fit semi-parametric model. Variable selection techniques are also introduced to the model. Since an insurance company has a large amount of client information, this paper focuses on the essential part. Too much information will not only distract attention but also waste time and effort on collection and storage. We would like to focus on the most important features or behaviors that influence the clients' decision. The techniques used in this paper include Lasso (Tibshirani, 1997), SCAD (Fan and Li, 2001), and MCP (Zhang, 2010). Oracle properties of variable selection via nonconcave penalized likelihood have been proved by Fan and Li (2002). These regularizations are accompanied by types of penalty terms and are widely recognized in variable selection.

3. Cox regression model

3.1. Definitions

In survival analysis, there are two essential concepts: survival probability and hazard. Let T be a nonnegative random variable denoting the failure time of an individual. The survival probability of time t, written as $S(t)$, represents the probability of an individual living to time t:

$$S(t) = P(T > t), \quad 0 < t < \infty \tag{1}$$

The hazard function $h(t)$ specifies the instantaneous rate at which failure happens to items that have survived at time t.

$$h(t) = \lim_{\Delta t \to 0^+} P(t \le T < t + \Delta t | T \ge t)/\Delta t \tag{2}$$

The hazard function and the survival probability both fully determine the distribution of t, and they can be converted to each other as:

$$h(t) = -[dS(t)/dt]/S(t) = -d \log S(t)/dt \tag{3}$$

$$S(t) = \exp[-\int_0^t h(s)ds] \tag{4}$$

In 1972, Cox (1972) proposed a semi-parametric regression model, the Cox regression model (a.k.a. the proportional hazards model) and since then, it has been the most widely used multi-factor model in

survival analysis. Similar to parametric models, it has parameters to estimate, while it does not depend on a specific distribution. Under the continuous assumption, the hazard function of the Cox model is:

$$h(t) = h_0(t)\exp(\beta X) \tag{5}$$

where $h_0(t)$ is an arbitrary unspecified baseline hazard function, and X is a vector of derived covariates that indicate certain features we collect about the very item. Hazard function is an important concept in survival analysis, which models the chance that an item fails at time t, conditional on its survival until time t (that is to say, its survival time $T \geq t$). The corresponding survival function has the form of

$$S(t, X) = S_0(t)^{\exp(\beta X)} \tag{6}$$

where $S_0(t) = \exp[-\int_0^t h_0(\mu)d(\mu)]$ is the arbitrary baseline survival function corresponding to $X = 0$.

The Eq. (1) can be converted to $\ln[f(t)/f_0(t)] = \beta X$. Thus, for each individual, the ratio of its hazard function and baseline hazard function is time-irrelevant, which is denoted as the "Proportional Hazard (PH) Assumption". In this paper, we focus on the cases in which the features do not vary with time. Thus, the PH Assumption holds. Next, we will introduce partial likelihood, which is often applied to infer β.

3.2. Inference through partial likelihood estimation

Suppose that a data set comprises a vector of observations Y, which has a density function $f(y; \beta, \theta)$. The β denotes the parameters of interest, while θ denotes the parameters of nuisance variable. The information of Y can be transformed into a new set of variables $A_1, B_1, A_2, B_2, \ldots, A_m, B_m$. Here, A_j and B_j contain two pieces of information that have no overlapping. Then, the joint density is obtained as:

$$\begin{aligned} f(A^{(m)}, B^{(m)}) &= \prod_{j=1}^{m} f(b_j|a^{(j-1)}, b^{(j-1)}; \beta, \theta) \prod_{j=1}^{m} f(a_j|a^{(j-1)}, b^{(j)}; \beta) \\ A^{(m)} &= (A_1, A_2, \ldots, A_m) \\ B^{(m)} &= (B_1, B_2, \ldots, B_m) \end{aligned} \tag{7}$$

Here, the second term $\prod_{j=1}^{m} f(a_j|a^{(j-1)}, b^{(j)}; \beta)$ only contains parameter β and is named as the partial likelihood of β, based on A_j in the sequence $\{A_j, B_j\}$. We denote it as $L(\beta)$. Different from the ordinary likelihood function, $L(\beta)$ cannot be interpreted explicitly in probabilities, either by marginal probabilities or conditional probabilities. Nevertheless, it is used in many large-sample instances for which asymptotic properties and properties of likelihood method hold. Here we use partial likelihood estimation to infer β in the Cox model.

Consider the hazard function $f(t) = f_0(t)\exp(\beta X)$ of the Cox Model under independent right censorship. For simplicity, we assume that there are no tied failure times. Suppose that the sample consists of k uncensored failure time $t_1 < t_2 < \cdots < t_k$ and that other $n-k$ individuals are right censored. Let A_j represent the failure of the jth item in the time interval $[t_j, t_j + dt_j)$ and B_j represent the censoring information in the time interval $[t_{j-1}, t_j)$, plus the knowledge that one item fails in $[t_j, t_j + dt_j)$. Then the partial likelihood associated with the jth item can be written as

$$L_j(\beta) = f(a_j|b^{(j)}, a^{(j-1)}; \beta) \tag{8}$$

The interpretation of $L_j(\beta)$ is very intuitive. It specifies the conditional probabilities of $j's$ failure in $[t_j, t_j + dt_j)$, given the information of all censorship and failures before t_j^- and the information that one item fails in $[t_j, t_j + dt_j)$. Let $R(t)$ denote the set of items at risk of failure at time t (which means they have not failed or been censored just prior to time t), we rewrite (3) as

$$L_j(\beta) = \frac{f(t_j; X_j)}{\sum_{l \in R(t_j)} f(t_j; X_l)} \tag{9}$$

The partial likelihood function of β is simply the product of (4). Under the Cox model it has the following form

$$L(\beta) = \prod_{i=1}^{k} \frac{f(t_i; X_i)}{\sum_{l \in R(t_i)} f(t_i; X_l)} = \prod_{i=1}^{k} \frac{\exp(\beta X_i)}{\sum_{l \in R(t_i)} \exp(\beta X_l)} \tag{10}$$

which only contains the target parameter β, and we obtain β by maximize $\ln L(\beta)$

$$\tilde{\beta} = \arg\max \ln L(\beta) = \arg\max\ l(\beta) = \arg\max_{\beta} \{\ln \prod_{i=1}^{k} \frac{\exp(\beta X_i)}{\sum_{l \in R(t_i)} \exp(\beta X_l)}\} \tag{11}$$

The Cox regression model has been widely used in survival analysis owing to its simplicity and negligence of real distribution. Nevertheless, the model requires n>p and that covariates should not be significantly related, which causes a major problem in practice. In addition, it will require much effort and time to collect all information on these variables, and data storage will inevitably be a problem. Therefore, variable selection should be considered. Several methods of doing so will be introduced in this paper.

4. Variable selection

4.1. Lasso regularization

Lasso penalty, since being raised in the 1990s, has become a popular regularization tool in various regression models. It uses L1-norm to constrain parameters and force the small ones to shrink to zero. The penalty is formed as:

$$p_\lambda(|\beta|) = \lambda|\beta| \tag{12}$$

Lasso brings sparsity effectively and has been proven to be variable selection-consistent (Meinshausen and Buhlmann, 2006; Tropp, 2006). However, the penalty increases linearly with the magnitude of its argument, which causes biases in large regression coefficients. To overcome the drawbacks of Lasso, many new regularization methods have been proposed. Two of them are presented in the next subsections.

4.2. Smoothly Clipped Absolute Deviation (SCAD)

Fan and Li (2001) stated three necessary properties of an estimator with a good penalty function: unbiasedness, sparsity, and continuity. As mentioned previously, the Lasso penalty has problems in terms of unbiasedness, and other methods such as the hard thresholding penalty result in estimators that are not continuous, which can cause instability in model prediction. In that paper, they developed SCAD, as an alternative variable selection method. They provided the specific conditions for penalization, under which the variable selection consistency holds, and large coefficients are unbiased.

The SCAD penalty function is formed as

$$p_{\lambda,\gamma}(|\beta|) = \begin{cases} \lambda|\beta| & if\ |\beta| \leq \lambda, \\ \frac{2\gamma\lambda|\beta| - |\beta|^2 - \lambda^2}{2(\gamma-1)} & if\ \lambda < |\beta| \leq \gamma\lambda, \\ \frac{\lambda^2(\gamma+1)}{2} & if\ |\beta| > \gamma\lambda. \end{cases} \tag{13}$$

for some $\gamma > 2$. The corresponding first-order derivative is

$$p'_{\lambda,\gamma}(\beta) = \begin{cases} \lambda & if\ |\beta| \leq \lambda, \\ \frac{\gamma\lambda - |\beta|}{\gamma - 1} & if\ \lambda < |\beta| \leq \gamma\lambda, \\ 0 & if\ |\beta| > \gamma\lambda. \end{cases} \tag{14}$$

According to Fan, SCAD performs well for variable selection when $\gamma = 3.7$. Thus, we conduct the method with this choice.

4.3. Minimax Concave Penalty (MCP)

Zhang (2010) proposed a new penalty function, the Minimax Concave Penalty (MCP), which is fast, nearly unbiased, and selection-accurate. The method provides selection consistency under the sparse Riesz condition, including the case of $p \gg n > d^0 \to \infty$. Similar to SCAD, the MCP has a form of a quadratic spline function while it has one knot

$$p_{\lambda,\gamma}(|\beta|) = \begin{cases} \lambda|\beta| - \frac{|\beta|^2}{2\gamma} & if\ |\beta| \le \gamma\lambda, \\ \frac{1}{2}\lambda^2\gamma & if\ |\beta| > \gamma\lambda. \end{cases} \tag{15}$$

where $\gamma > 0$ and $\lambda > 0$ are two tuning parameters. It has the first-order derivative as

$$p'_{\lambda,\gamma}(\beta) = \begin{cases} \lambda - \frac{|\beta|}{\gamma} & if\ |\beta| \le \gamma\lambda, \\ 0 & if\ |\beta| > \gamma\lambda. \end{cases} \tag{16}$$

4.4. Penalized partial likelihood

The preceding sections demonstrate the inference of β in the Cox model and variable selection through different methods, which provide a natural hint about our client attrition problem. Supposing there are k features we know about every client, we can obtain the penalized likelihood function

$$\sum_{i=1}^{n} \ln \frac{\exp(\beta X_i)}{\sum_{l\in R(t_i)} \exp(\beta X_l)} - n\sum_{j=1}^{k} p_\lambda(|\beta_j|) \tag{17}$$

Maximizing the penalized likelihood function is equivalent to minimizing

$$-\sum_{i=1}^{n} \ln \frac{\exp(\beta X_i)}{\sum_{l\in R(t_i)} \exp(\beta X_l)} + n\sum_{j=1}^{k} p_\lambda(|\beta_j|) \tag{18}$$

Using $p_\lambda(|\beta|)$ given in 3.1–3.3, we can find the essential features that influence clients' insurance renewal decisions, along with their coefficients.

5. Simulation and applications

5.1. Simulation

To implement the models given in previous sections, we conduct a simulation study. We simulate 100 data sets, with each consisting of $n = 100$ samples under the Cox Model with a hazard function as follows

$$h(t) = \exp(\beta X) \tag{19}$$

Here, $\beta = (0.3, 0, 0, 0.6, 0, 0, 0.5, 0, 0, 1)$ and X is a vector of 10 elements with x_i marginally following normal distributions. To ensemble the reality, we set the variables correlated to each other and $\rho(x_i, x_j) = \rho^{|i-j|}$. Here, we set $\rho = 0.5$. Also, we consider the independent censoring problem and let the censoring time follow an exponential distribution with a given mean $U\exp(\beta X)$. If the censoring time is shorter than the failure time, we assume that the sample is censored. U is simulated in every set with a uniform distribution over $[0.2, 0.5]$, which controls the censoring size, mainly in 20%−40%. The average coefficients of the 10 variables under three penalized Cox Models are showed in Fig. 1. We can see that the average coefficients of β_1, β_4, β_7 and β_{10} are significantly higher than those of the rest, which is in accordance with the fact.

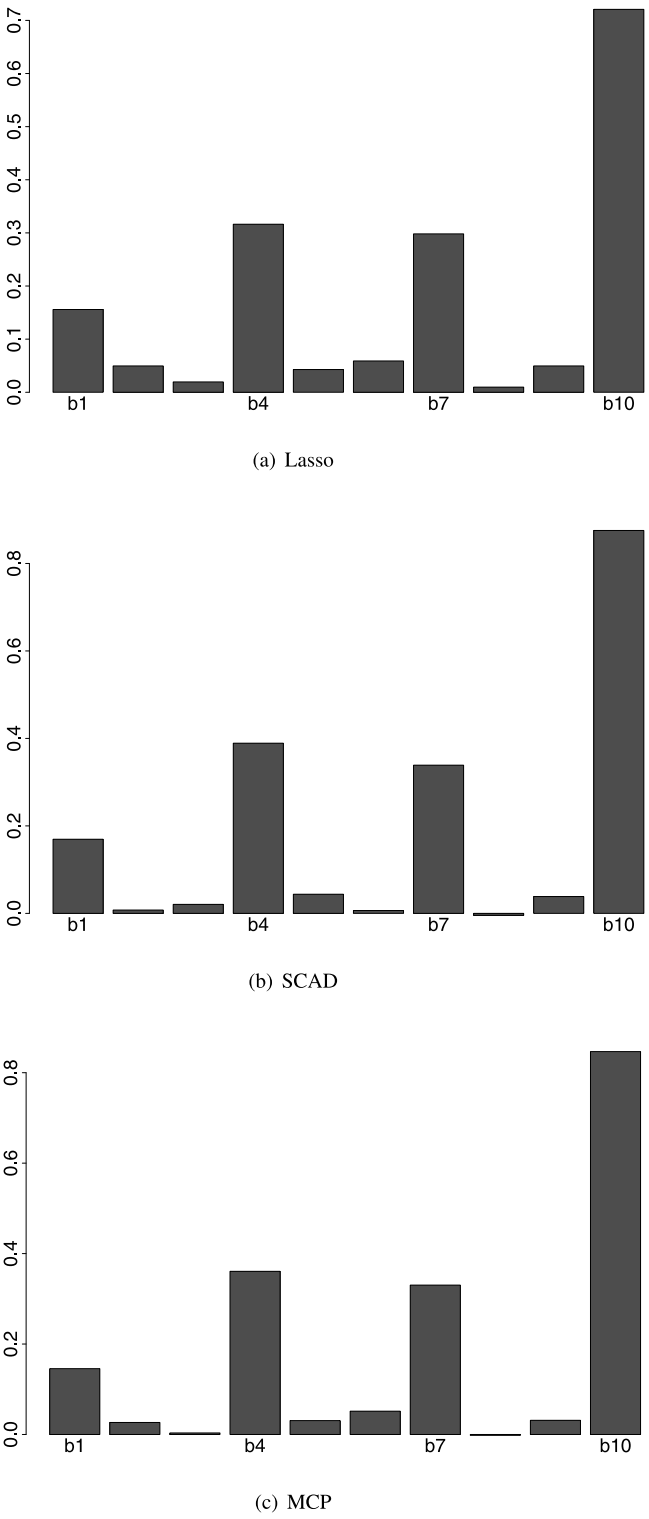


Fig. 1. Coefficients of the three penalized Cox Models. β_1, β_4, β_7 and β_{10} are set nonzero.

5.2. Data

The dataset is provided by a major insurance company in China. We set July 1, 2016 as the stopping time ("S" for short) and randomly retrieve the insurance renewal information about 20,960 clients of the preceding five years. Each client has 47 related features, including 45 numerical variables and two string variables. The features can be divided into four classes, and we provide some examples in Table 1.

This experiment involves studying the client churn rate. Therefore, we set the contract termination as the "failure" in the Cox Model. Accordingly, clients that retain the contract until the end of the experiment are regarded as censored. For convenience, we use the terminology shown in Table 2.

Thus, the dataset can be interpreted as $(T_i, I_i, X_i), i = 1, 2, \ldots, 20\,960$. In the data preparation session, we take three steps. First, for those sting-type variables, we arrange them into proper classes (if necessary)

Table 1
Features collected about every client.

Classes	Features
Personal information	Gender, age, education level, marital status, …
Business profile	Total assets, total liabilities, risk-averse tendency, investment experience, …
Insurance behavior	Average contract duration, number of car contracts held at ST, total payments for car contracts at ST, …
Loyalty	Frequency of opening car insurance company's website, interval from the last time opening car insurance company's website, …

Table 2
Terminology of the experiment.

Symbols	Definitions
T_i	The contract retention time of client i.
I_i	An indication function: $I_i = 0$, client i keep renewing the contract till the end; $I_i = 1$, client i terminate the insurance before the stopping time.
X^j	The jth feature
X_i	The vector of features about client i

Table 3
The hyperparameters of Lasso, SCAD, and MCP penalties in the experiment. The values of γ are chosen as the originators suggest, and the values of λ are decided by attaining the smallest cross-validation error.

	γ	λ	Cross validation error
Lasso		0.02	12.62
SCAD	3.70	0.22	11.63
MCP	3.00	0.36	15.84

and use instrumental variables in the regression. For instance, when it comes to marriage status, there are four types: single, married, divorced, and other. To avoid multicollinearity, we set three instrumental variables for "single", "married" and "divorced", and those three variables being zero indicate the status "other". Hence, we have 49 variables in total, including four instrumental variables. Second, for those missing data, we fill the blanks with the average of the rest. There are few missing values, therefore, the model should not be affected. Third, we find that there are some variables that have vast domains and are seriously right-skewed. For instance, the variable "total assets" varies from 0 to 70 million, and the 99% quantile is 1.15 million. For those variables, we replace the value of the largest 1% portion with the 99% quantile. Third, we standardize the data prior to modeling.

We randomly divide the sample set into one training set and one test set. The training set consists of 16,768 samples, accounting for 80% of the total.

5.3. Regression with Lasso, SCAD, and MCP

In this section, we apply three penalized Cox models to the training set. The value of tuning parameter λ can make a huge difference in the variable selection, and a larger value can create greater sparsity. Here, we use tenfold cross-validation and choose the value of $lambda$, 0.02, which minimizes the cross-validation error. The Lasso penalty selects eight variables. We present the selected variables and regression coefficients in Table 4.

We proceed with the same work on SCAD and MCP penalties. As for the hyperparameter γ, we use $\gamma = 3.70$ and $\gamma = 3.00$, respectively, as the originator suggests. Following the work on the Lasso penalty, we use tenfold cross-validation and let the cross-validation error lead us to the best value of λ. The results of the hyperparameter selection on three penalties are shown in Table 3. As shown in Table 4, the SCAD penalty selects two variables, while the MCP penalty selects one.

The selected variables have substantial practical meanings. For instance, X^{21} is defined as the number of car insurance policies held at the same time. The coefficients being positive indicates that people who hold more car insurance policies are more likely to leave a company than the rest. This tendency is consistent with the practical experience gained by companies. These clients have a relatively larger scope of choices, and there is a greater chance of them switching options. There is another possible cause. We guess they probably have more vehicles and are consequently followed by more insurance salespeople, which

Table 4
Lasso, SCAD, and MCP penalized Cox models that select eight, two, and one variables, respectively.

X^j	coef	exp(coef)	se(coef)	z	Pr(> |z|)
Lasso					
X^1	0.035	1.035	0.001	57.343	<0.001
X^4	0.292	1.340	0.062	4.696	<0.001
X^5	0.000	1.000	0.000	11.362	<0.001
X^6	0.197	1.218	0.018	10.798	<0.001
X^9	−0.000	1.000	0.000	−3.052	0.002
X^{17}	−1.092	0.336	0.024	−46.280	<0.001
X^{18}	−0.467	0.627	0.020	−23.658	<0.001
X^{21}	0.164	1.178	0.043	3.801	<0.002
SCAD					
X^1	0.036	1.036	0.001	62.830	<0.001
X^{17}	−0.992	0.371	0.017	−57.890	<0.001
MCP					
X^{17}	−1.147	0.318	0.015	−76.590	<0.001

may lead them to try other companies' offers. By contrast, X^{17}, with the definition of the times the client purchases a company's car compulsory insurance, has negative coefficients. This makes sense because this variable exhibits the repurchase behavior of the client, and a larger value shows a higher level of loyalty. It is selected by all three penalized models, and, under the MCP method, it is the only variable that is not eliminated.

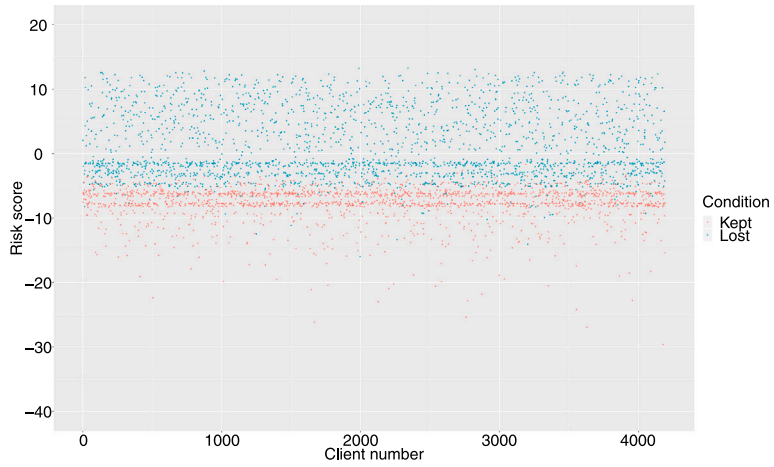
5.4. Risk score

The models provide us with an intuitive way to obtain a risk score for each client to measure the leaving possibility. From the previous introduction, we know that the hazard function of a client is $h(t) = h_0(t)\exp(\beta X)$. The baseline function $h_0(t)$ is identical for every client, so we can use βX to compare them. Here, we define the risk score of a client i leaving the company:

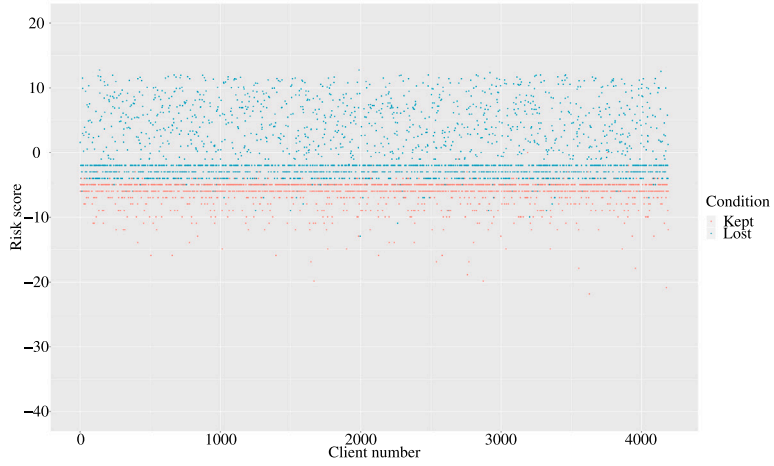
$$RS_i = \beta X_i \tag{20}$$

Applying the β we obtained from the Cox Model, we calculate the risk score of all clients in the test set. We present the result in Fig. 2, in which the lost clients (blue points) and the retained clients (red points) are roughly separated into two parts. Table 3 shows the first 20 clients' risk scores, and we can see that lost clients tend to have higher risk scores than the opposite (see Table 5).

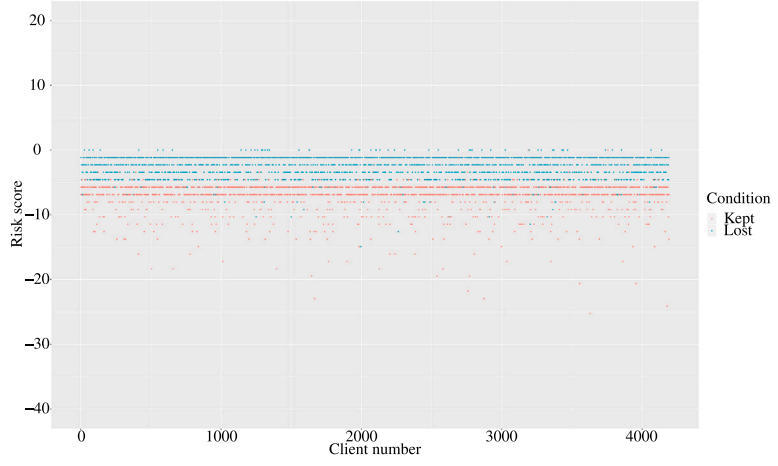
It should be noted that under the MCP penalty, the points nearly cumulate in horizontal lines because there is only one variable left. This occurs to the points under the SCAD penalty too, for that many X_i^1 equal zero. The evidence we obtain in the previous paragraphs indicates that the risk score with a specified threshold can be a classifier, helping to determine if a client is likely to leave a company. One of the effective evaluation criteria for classification models is the Area Under Receiver Operating Characteristic Curve (AUROC). The ROC curve is a graphical



(a) Lasso



(b) SCAD



(c) MCP

Fig. 2. Risk scores of all clients in the test set, under (a) Lasso, (b) SCAD, and (c) MCP penalties, respectively. It is shown that the scores of lost clients are generally higher than those of retained clients. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

plot with two coordinates: a True Positive Rate (TPR) on the horizontal axis and a False Positive Rate (FPR) on the vertical. The plot illustrates the model's discrimination ability as the threshold varies. The larger the AUROC, the better the classification performance offered by the model. In Fig. 3, we illustrate the ROC curves of risk-score classifiers under Lasso, SCAD, and MCP penalties, respectively.

ROC curves in Fig. 3 show that all three classifiers under different penalties can divide clients effectively. The plots of SCAD and MCP look similar because when $X^1 = 0$, the Lasso penalized model has only one variable X^{17}, which is the same as the SCAD penalized model. From Fig. 2, we know that clients with $X^1 = 0$ are very "controversial". AUROCs for three classifiers with Lasso, SCAD, and MCP penalties

Table 5
Risk scores of the lost clients are generally higher than the others.

Client number	Condition	Risk score		
		Lasso	SCAD	MCP
17058	Kept	−6.63	−4.96	−5.73
13970	Lost	6.26	5.92	−1.15
10458	Lost	−2.77	−2.98	−3.44
16532	Lost	11.06	10.87	−1.15
17292	Kept	−9.08	−6.95	−8.03
13311	Lost	−1.26	−1.98	−2.29
7274	Kept	−7.45	−6.95	−8.03
20858	Kept	−6.22	−4.96	−5.73
11752	Kept	−8.00	−5.95	−6.88
7400	Kept	−6.24	−4.96	−5.73
15363	Kept	−6.47	−4.96	−5.73
7840	Kept	−6.38	−4.96	−5.73
14358	Lost	−2.42	−2.98	−3.44
6762	Kept	−8.05	−5.95	−6.88
17113	Kept	−6.15	−4.96	−5.73
13183	Kept	−11.52	−8.93	−10.32
2704	Lost	−3.38	−2.98	−3.44
8348	Lost	−1.41	−1.98	−2.29
5668	Kept	−10.74	−8.93	−10.32
2369	Lost	7.07	6.56	−1.15

Table 6
Information of the three models collected during the experiment.

	Variable number	R^2	Concordance	AUROC
Lasso	8	0.93	0.96	0.0.9791
SCAD	2	0.91	0.96	0.9767
MCP	1	0.79	0.92	0.9762

equal 0.9791, 0.9767, and 0.9762, respectively, which shows that these classifiers are effective tools for identifying potential lost clients.

5.5. Model comparison

Previous subsections record the experiment and give the results with some explanations. In this part, we collect some complementary information and make a thorough comparison of the models. The information is shown in Table 6, and MCP creates the greatest sparsity.

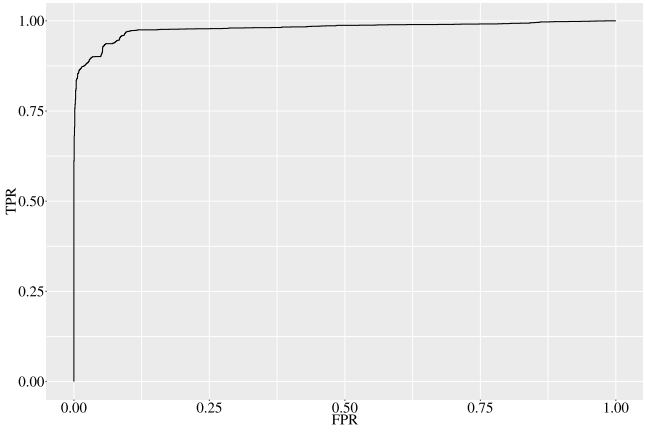
Concordance, which is an important index in survival analysis, measures the prediction power of the applied model. It is calculated in three steps:

- Step1: List all possible pairs in the sample set and eliminate the pairs in which two clients are both kept (we do not know who in the pair will leave the company first). Set the number of pairs as M.
- Step2: Find the pairs in which the client who is predicted to stay longer actually stays longer. Set the number of these pairs as K.
- Step3: The concordance index is defined as M/K.

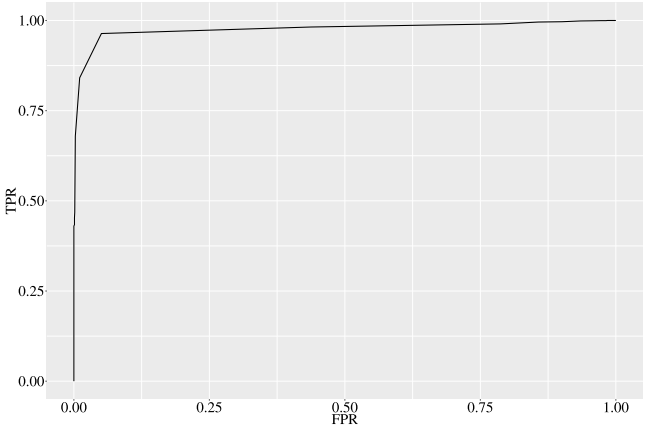
We demonstrate the results shown in Table 6 in three aspects:

- $Sparsity$: As expected, MCP creates the greatest sparsity, and the Lasso method still holds on to nine variables. More variables means more effort on data collection and computation, which will create more inconvenience in practice.
- $Fitness$: Lasso strikes the best fitness, considering the highest R^2. However, SCAD also performs well with many fewer variables than Lasso. MCP has the lowest R^2, in which only one variable remains, and too much information is lost.
- $Classification\ accuracy$: The three models perform just as well. We think this is because X^{17} is a typical variable of loyalty, which plays an important role in clients staying with the company.

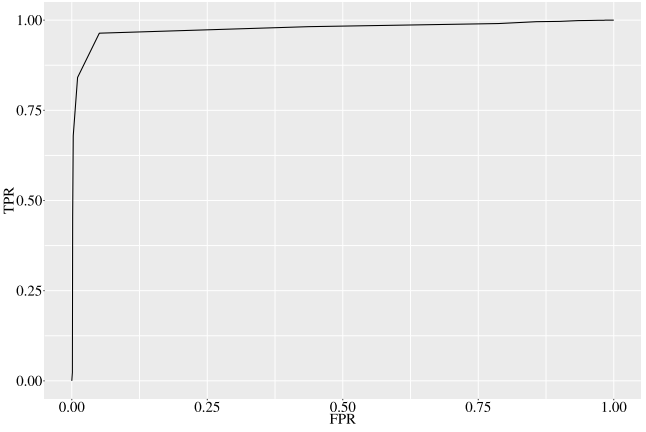
In conclusion, we think SCAD is the best penalty in our experiment. In the next part, we will introduce the idea of the dynamic threshold.



(a) Lasso



(b) SCAD



(c) MCP

Fig. 3. ROC curves of Lasso (a), SCAD (b), and MCP (c) penalized risk-score classifiers. AUROC=0.9767, 0.9767, and 0.9762 for each, which shows they are good classifiers.

5.6. Dynamic threshold for survival probability

Through the research in the previous section, when using the survival analysis method to solve the client churn problem, the author often faces a problem after obtaining the "survival" probability of the client, that is, how to judge whether the client will actually leave. The general approach is to give a fixed threshold based on past experience. However, in reality, the client's status is dynamic. After signing the contract, a client can change his or her expectations and dependence on the company with time. When using a fixed threshold, we assume that

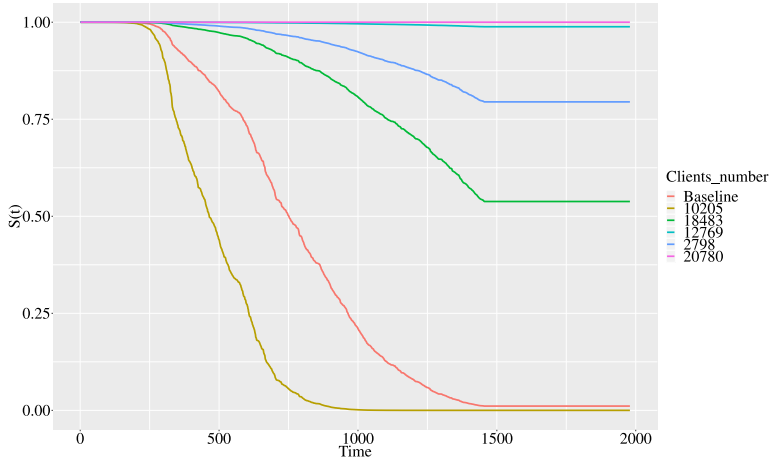


Fig. 4. The baseline and five randomly selected clients' survival curves under the SCAD penalized model. Clients 2798, 10 205, and 18 483 have actually lost, and correspondingly, they have lower "survival" probabilities compared with the other two.

the client's expectations and dependence on the company are constant. Therefore, we propose a dynamic threshold method based on the real business case.

As we learn from business, client relationship maintenance is important. The cost of acquiring one new client is much higher than that of maintaining one, and meanwhile, the profit from a new client is much smaller than the loss of losing a regular one. Hence, we should attach importance to recalling value, while promising a high precision value. Here, we first give the formula of *Precision* and *Recall*:

$$Precision = \frac{\{Actually\ lost\ client\} \cap \{Predicted\ lost\ client\}}{\{Predicted\ lost\ client\}} \tag{21}$$

$$Recall = \frac{\{Actually\ lost\ client\} \cap \{Predicted\ lost\ client\}}{\{Actually\ lost\ client\}} \tag{22}$$

Low *Precision* can create a great waste of human and material resources, while low *Recall* reflects that many lost clients have not been taken into consideration. We randomly select five clients to demonstrate the definitions. Under the SCAD penalized Cox model, the baseline and five clients' survival curves are shown in Fig. 4. The survival curves of clients 2798, 10205, and 18483 lie beneath the curves of the other two, which corresponds to the fact that they have left the company.

For instance, on day 1000, if the threshold is set at 0.5, we can get $Precision = 1/1 = 1$ and $Recall = 1/3 = 0.33$; if the threshold is set at 0.88 on that day, *precision* will be 1, and *Recall* will be 0.67. On day 1500, if the threshold is set at 0.88, *precision* will be 1, and *Recall* will be 1. Thus, different days and thresholds we will produce different precision and recall. Our target is to build a dynamic threshold which balances precision and recall.

Here, we introduce the way to obtain a dynamic threshold. Under the SCAD penalized Cox model, we first draw the precision curve and recall curve over a specified range of thresholds on a certain time point and record the threshold with which the two curves intersect. Then, we plot the change of recorded thresholds over time. This method balances prediction accuracy and actual recall rate. Next, we provide an example to demonstrate how it is to be done. In Fig. 5, we show the precision curves and recall curves of day 1000 and day 1500, over the threshold range $[0, 1]$ with varying steps equal to 0.001.

As shown in Fig. 5, the curves are actually stepwise, so they do not necessarily have real intersections. On day 1500, when the threshold comes up to 0.919, *Precision* descends from 0.992 to 0.961, while *Recall* jumps to 0.965 from 0.842; at this very point, recall takes the place of precision and becomes dominant. For balance, we record 0.919 as our desired threshold. We go through other days in the same way. The thresholds over days are presented in Fig. 6.

Table 7
In both the training set and test set, *Precision* and *Recall* of the dynamic threshold are considerably high.

	Training set	Test set
Precision	0.921	0.920
Recall	0.971	0.970

We apply the dynamic threshold back to the training set and test set. The average *Precision* and *Recall* for the two sets is shown in Table 7. We can see that in the test set, the dynamic threshold performs as well as it does in the training set. We observed that with smaller varying steps when building the dynamic threshold, performance in both *Precision* and *Recall* will improve.

One significant use of the dynamic threshold is that companies can manage clients' contracts over time. After collecting a client's features, a company can draw his or her survival curve and predict the survival probability at a time point. The company can always compare the probability with the dynamic threshold, and if the probability comes up short, the company should pay attention to this client. Meanwhile, if two clients are both potentially leaving, the one with a smaller probability definitely needs more attention. *Precision* and *Recall* index will be high, which lends strong support to this plan. If time-varying features are considered, the dynamic threshold can be of greater help, but that is beyond the purpose of this paper.

6. Conclusion and future work

Nowadays, insurance companies are looking for new ways to maintain the client relationship, and the churn prediction model can be seen as one of the alternatives. An effective model can accurately predict the client churn trends, and rank the database of users with the possibility of churn. Based on this, insurance companies can implement retention measures for users with high churn rate in advance to avoid or reduce the client churn rate.

This paper proposes a new model, i.e., combined the Cox model with variable penalties (Lasso, SCAD, and MCP) to predict a client's leaving and distinguish regular clients for vehicle insurance, through knowledge of survival analysis. In essence, this can also be understood as a data mining technology, which aims to help insurance companies to dynamically maintain customer relationships and reduce the number of customer churn. This paper investigates the use of survival analysis in client relationship management, focusing on predicting whether a client will leave a company. We collect the complementary information and make a thorough comparison of these models. As expected, MCP creates the greatest sparsity, and Lasso strikes the best fitness,

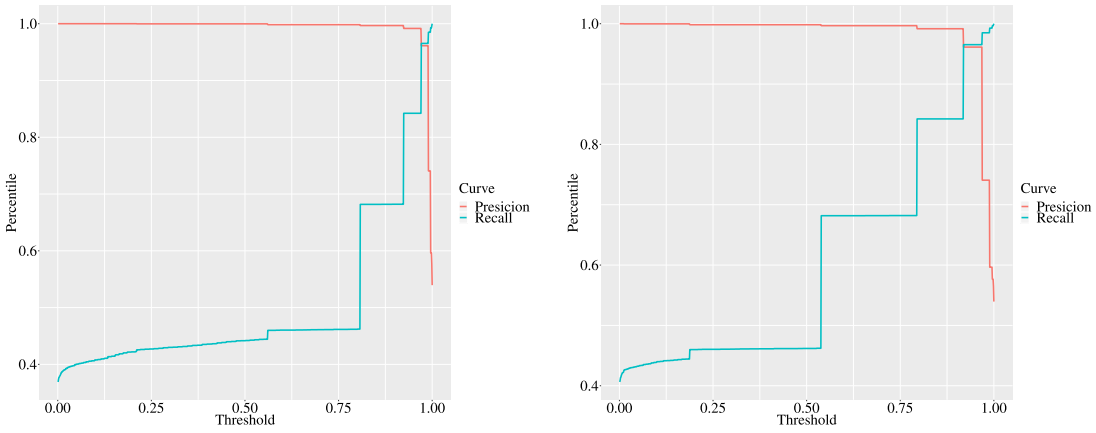


Fig. 5. Precision curves and recall curves over thresholds in range $[0, 1]$, on day 1000 (a) and day 1500 (b). We are interested in the thresholds of intersections because they balance the precision and recall at the same time.

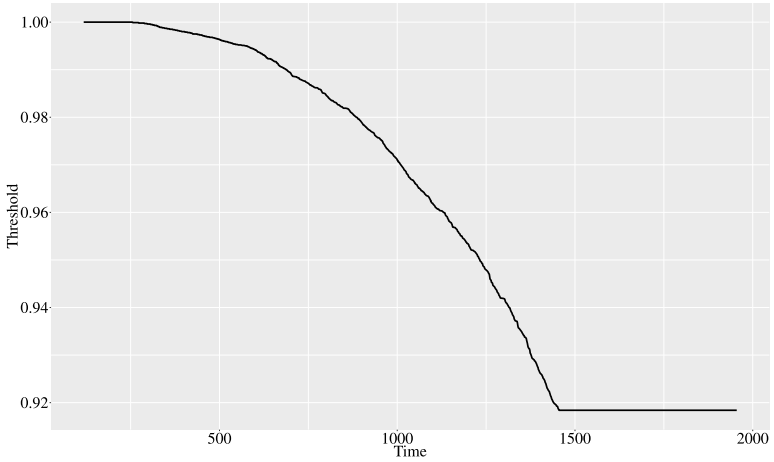


Fig. 6. Dynamic threshold curve based on the training set.

considering the highest R^2. However, SCAD also performs well with fewer variables than Lasso. MCP has the lowest R^2, in which only one variable remains, and too much information is lost. The AUROC of the model on the test set is considerably high, which shows that the model can be quite helpful for helping a company to precisely distinguish potentially leaving clients from regular ones. Furthermore, we develop a dynamic threshold of clients' staying probability over time, by balancing the classification of *Precision* and *Recall*. At any time, one can always compare a client's staying probability with this threshold and determine if the client needs particular attention. After collecting a client's features, a company can draw his or her survival curve and predict the survival probability at a time point; the company can always compare the probability with the dynamic threshold, and if the probability comes up short, the company should pay attention to this client. Meanwhile, if two clients are both potentially leaving, the one with a smaller probability definitely needs more attention. *Precision* and *Recall* index will be high, which lends strong support to this plan.

One illuminating idea that emerged when processing our experiment is that we can add a time-varying effect to the dynamic threshold, and clients' staying curve should be modified when their features change. With these improvements, the dynamic threshold curve will have intersections with clients' staying curves, and the intersection will alert the company to the possibility that the client may leave. In this way, we can not only predict a client's leaving but his or her "dangerous" time. Obviously, this can help a company manage its clients' relationships more flexibly. Churn prediction data from insurance companies often have very complex characteristics, such as missing data, noise, etc. How to find suitable solutions for data with different characteristics, and summarize the research on customer churn prediction oriented to noise and missing data Paradigm will be left for future work.

CRediT authorship contribution statement

Yan Chen: Writing – review & editing, Project administration, Methodology, Conceptualization. **Lei Zhang:** Writing – original draft, Visualization, Software, Formal analysis, Data curation. **Yulu Zhao:** Writing – original draft, Visualization, Methodology, Data curation. **Bing Xu:** Writing – review & editing, Supervision, Methodology, Formal analysis.

Data availability

Data will be made available on request.

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